Count Occurrences of a Target in an Array

Problem statement

Given an integer array of length N and a target K, find how many elements equal K.

Input format

First line: N and K. Second line: N whitespace-separated integers.

Output format

Print the count as one integer on its own line.

Constraints

- 1 <= N <= 1000
- -10000 <= K, A[i] <= 10000

Sample input and output

Sample input	Expected output
`6 2` `2 3 2 -1 2 0`	`3`

Important boundary conditions

The target may be negative or zero; duplicates are allowed.

Implementation expectations

O(N) time and O(1) extra space are possible.

Submission requirements

Submit exactly one .c file. Use stdin and stdout without interactive prompts. Compile with GCC in C11 mode. Avoid undefined behavior, release dynamically allocated memory, and print precisely the required output. Follow the stated input constraints.

Teacher resources (not part of student handout)

A reference oracle, five sample student submissions, assignment-specific rules.json, and 12 verified tests are supplied separately in the instructor test pack. Do not distribute the reference oracle or deliberately faulty variants as student solutions.